NEET UG PHYSICS SYLLABUS

Systems of Particles and Rotational Motion

Complete Exam-Oriented Tutorial: Centre of Mass, Torque, Angular Momentum, Equilibrium, and Fixed-Axis Rotational Dynamics

1. Learning Objectives & Chapter Overview

What you will master in this tutorial:

- Distinguish between a point particle, an extended system of particles, and an ideal **rigid body**, and classify rigid-body motions (pure translation, pure rotation, and combined motion).
- Calculate the **centre of mass (CM)** for discrete particle systems, continuous homogeneous bodies, and composite/cavity laminas.
- Analyze the **motion of the centre of mass**, total linear momentum, and conservation of linear momentum under zero external force.
- Apply the **vector (cross) product** to compute **torque (moment of force)** and **angular momentum**, and relate angular velocity to linear velocity via $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$.
- Apply the conditions of **mechanical equilibrium** (translational and rotational), the principle of moments, couples, and the concept of **centre of gravity**.
- Understand **moment of inertia** and **radius of gyration** for standard bodies, and solve fixed-axis rotational kinematics, dynamics ($\tau = I\alpha$), work-energy, power, and angular momentum conservation ($I\omega = \text{constant}$) problems.

2. Extended Bodies, Rigid Bodies, and Types of Motion

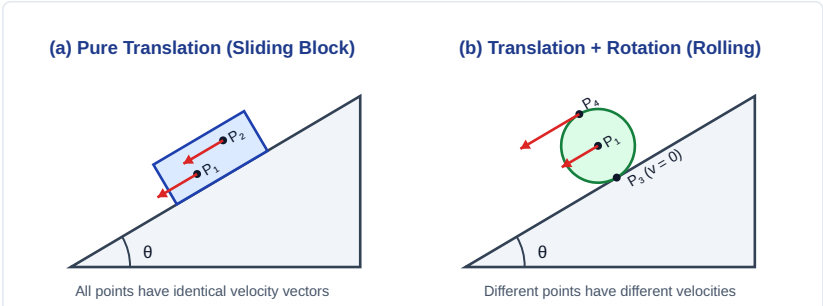
In elementary mechanics, bodies of finite size are often modeled as single particles (point masses having no size). However, when dealing with real extended bodies, the particle model is frequently inadequate. An extended body is fundamentally a **system of particles**. A large class of problems involving extended objects—such as wheels, spinning tops, steel beams, molecules, and planets—can be solved by treating them as **rigid bodies**.

Definition of a Rigid Body: Ideally, a rigid body is a body with a perfectly definite and unchanging shape. The distances between all pairs of particles of such a body do not change, even under the influence of forces. (No real body is truly rigid because all real bodies deform slightly under force, but in many practical situations these deformations—warping, bending, or vibrating—are negligible.)

2.1 What Kind of Motion Can a Rigid Body Have?

Depending on how a rigid body is constrained, its motion falls into distinct categories:

1. **Pure Translational Motion:** Consider a rectangular block sliding down an inclined plane without any sidewise movement. All particles of the block move together and have the exact same velocity at any given instant of time. The orientation of a line segment joining any two points within the body remains unchanged throughout the motion.
2. **Pure Rotational Motion about a Fixed Axis:** If a rigid body is constrained by fixing it along a straight line (such as a ceiling fan, a potter's wheel, a giant wheel, or a merry-go-round), it cannot undergo translational motion. Its only possible motion is rotation about that fixed straight line, called the **axis of rotation**. In rotation about a fixed axis, every particle of the rigid body moves in a circle that lies in a plane perpendicular to the axis and has its centre on the axis. Any particle lying directly on the fixed axis (distance $r = 0$) remains stationary while the body rotates.
3. **Rotation with One Point Fixed (Moving Axis):** In some cases of rotation, the axis is not fixed in space, though it always passes through a single fixed point.
   - *Spinning Top:* A top spinning in place without slipping has its tip (point of contact O with the ground) fixed. The axis of the top sweeps out a cone around the vertical line passing through O. This conical movement of the axis of rotation around the vertical is called **precession**.
   - *Oscillating Table or Pedestal Fan:* While the blades rotate about their central spindle, the spindle itself oscillates sidewise in a horizontal plane about a vertical pivot point O. Here, one point O (not a line) is fixed.
4. **Combined Translational and Rotational Motion:** Consider a solid cylinder rolling down an inclined plane. The cylinder shifts from the top to the bottom of the incline (translation), yet different particles of the cylinder have different velocities at the same instant. In fact, if the cylinder rolls without slipping, the instantaneous velocity of the point of contact with the incline is zero! Similarly, if a rigid object is thrown into the air while spinning, its centre of mass follows a parabolic translational trajectory while the orientation of the body continuously changes.



Tutorial Diagram 1: Comparison of pure translational motion of a sliding block and combined translational plus rotational motion of a rolling cylinder.

Fundamental Rule of Rigid Body Motion:

- The motion of a rigid body that is **not pivoted or fixed** in any way is either a *pure translation* or a *combination of translation and rotation*.

BOSONANCE

- The motion of a rigid body that is **pivoted or fixed** in some way is *rotation* (either about a fixed axis or a moving axis passing through the fixed point).

3. Centre of Mass of a System of Particles

3.1 Two-Particle and Multi-Particle Discrete Systems

Consider a system of two particles of masses m_1 and m_2 lying along the x-axis at distances x_1 and x_2 from the origin O. The **centre of mass (C)** of the system is located at a distance X from O, defined as the mass-weighted mean position:

$$X = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

If the two particles have equal mass ($m_1 = m_2 = m$), then $X = \frac{x_1 + x_2}{2}$, meaning the centre of mass lies exactly midway between them.

For a system of n particles of masses $m_1, m_2, \ldots, m_n$ distributed in three-dimensional space at positions (x_i, y_i, z_i) where total mass $M = \sum_{i=1}^{n} m_i$, the coordinates (X, Y, Z) of the centre of mass are:

$$X = \frac{\sum_{i=1}^{n} m_i x_i}{M}, \quad Y = \frac{\sum_{i=1}^{n} m_i y_i}{M}, \quad Z = \frac{\sum_{i=1}^{n} m_i z_i}{M}$$

For three particles of equal mass $m_1 = m_2 = m_3 = m$ not lying in a straight line, $X = \frac{x_1 + x_2 + x_3}{3}$ and $Y = \frac{y_1 + y_2 + y_3}{3}$, which coincides with the **centroid** of the triangle formed by the three particles.

Using position vectors $\mathbf{r}_i = x_i\hat{\mathbf{i}} + y_i\hat{\mathbf{j}} + z_i\hat{\mathbf{k}}$ and $\mathbf{R} = X\hat{\mathbf{i}} + Y\hat{\mathbf{j}} + Z\hat{\mathbf{k}}$, the three scalar equations combine into a single compact vector equation:

$$\mathbf{R} = \frac{\sum_{i=1}^{n} m_i \mathbf{r}_i}{M}$$

Important Property: If the origin of the coordinate system is chosen to be at the centre of mass itself ($\mathbf{R} = \mathbf{0}$), then the mass-weighted sum of position vectors vanishes: $\sum_{i=1}^{n} m_i \mathbf{r}_i = \mathbf{0}$.

3.2 Continuous Mass Distributions and Symmetry

For a rigid or continuous body containing a vast number of closely packed atoms, we subdivide the body into small mass elements Δm_i and take the limit as $\Delta m_i \to \mathrm{d}m$. The summation is replaced by integration over the volume/area/length of the body:

$$X = \frac{1}{M}\int x\,\mathrm{d}m, \quad Y = \frac{1}{M}\int y\,\mathrm{d}m, \quad Z = \frac{1}{M}\int z\,\mathrm{d}m \quad \Longleftrightarrow \quad \mathbf{R} = \frac{1}{M}\int \mathbf{r}\,\mathrm{d}m$$

If the centre of mass is chosen as the origin ($\mathbf{R} = \mathbf{0}$), then:

$$\int \mathbf{r}\,\mathrm{d}m = \mathbf{0} \quad \text{or} \quad \int x\,\mathrm{d}m = \int y\,\mathrm{d}m = \int z\,\mathrm{d}m = 0$$

Reflection Symmetry and Homogeneous Bodies: A body with uniformly distributed mass is called *homogeneous*. Consider a thin homogeneous rod along the x-axis with its geometric centre at the origin O. By reflection symmetry, for every mass element $\mathrm{d}m$ at position $+x$, there is an identical mass element $\mathrm{d}m$ at position $-x$. Their net contribution to $\int x\,\mathrm{d}m$ is $x\,\mathrm{d}m + (-x)\,\mathrm{d}m = 0$. By the same reflection symmetry argument ($(x, y, z) \leftrightarrow (-x, -y, -z)$), the centre of mass of homogeneous regular bodies—such as thin rods, thick rectangular or cylindrical bars, rings, circular discs, spheres, and cubes—always coincides with their **geometric centre**.

Note: The centre of mass does **not** necessarily lie inside the material of the body. For example, for a uniform circular ring or a hollow sphere, the centre of mass is at its geometric centre, where there is no material mass.

4. Motion of the Centre of Mass & Linear Momentum

Starting from the definition $M\mathbf{R} = m_1\mathbf{r}_1 + m_2\mathbf{r}_2 + \cdots + m_n\mathbf{r}_n$ and assuming the individual masses m_i remain constant in time, differentiating once with respect to time gives the velocity $\mathbf{V} = \frac{\mathrm{d}\mathbf{R}}{\mathrm{d}t}$ of the centre of mass:

$$M\mathbf{V} = m_1\mathbf{v}_1 + m_2\mathbf{v}_2 + \cdots + m_n\mathbf{v}_n = \sum_{i=1}^{n} m_i\mathbf{v}_i$$

Differentiating a second time with respect to time gives the acceleration $\mathbf{A} = \frac{\mathrm{d}\mathbf{V}}{\mathrm{d}t}$ of the centre of mass:

$$M\mathbf{A} = m_1\mathbf{a}_1 + m_2\mathbf{a}_2 + \cdots + m_n\mathbf{a}_n = \mathbf{F}_1 + \mathbf{F}_2 + \cdots + \mathbf{F}_n$$

Here, $\mathbf{F}_i$ is the total force on the ith particle, which includes both external forces (exerted by agencies outside the system) and internal forces (exerted by other particles within the system). By Newton's third law, internal forces between any pair of particles occur in equal and opposite pairs, so their vector sum over the entire system cancels out to zero ($\sum \mathbf{F}_{\text{int}} = \mathbf{0}$). Therefore, only external forces contribute:

$$M\mathbf{A} = \mathbf{F}_{\text{ext}}$$

Physical Significance: The centre of mass of a system of particles moves as if the entire mass M of the system were concentrated at the centre of mass and all the external forces $\mathbf{F}_{\text{ext}}$ were applied directly at that point. No knowledge of internal forces is required to determine the motion of the centre of mass!

4.1 Total Linear Momentum and its Conservation

The linear momentum of a single particle is $\mathbf{p} = m\mathbf{v}$. For a system of n particles, the total linear momentum $\mathbf{P}$ is the vector sum of the individual momenta:

BOSONANCE

$$\mathbf{P} = \mathbf{p}_1 + \mathbf{p}_2 + \cdots + \mathbf{p}_n = \sum_{i=1}^{n} m_i \mathbf{v}_i = M\mathbf{V}$$

Differentiating with respect to time gives Newton's second law for a system of particles:

$$\frac{\mathrm{d}\mathbf{P}}{\mathrm{d}t} = M\mathbf{A} = \mathbf{F}_{\text{ext}}$$

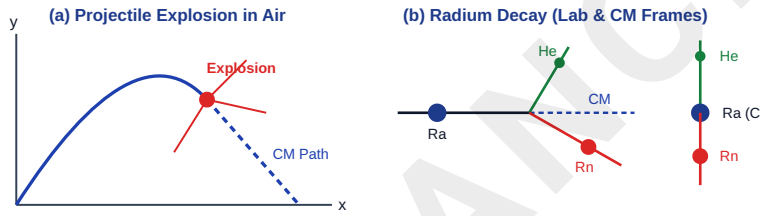
Law of Conservation of Linear Momentum: If the total external force acting on a system of particles is zero ($\mathbf{F}_{\text{ext}} = \mathbf{0}$), then $\frac{\mathrm{d}\mathbf{P}}{\mathrm{d}t} = \mathbf{0}$, which implies:

$$\mathbf{P} = M\mathbf{V} = \text{Constant} \quad \Longleftrightarrow \quad P_x = c_1, \;\; P_y = c_2, \;\; P_z = c_3$$

Thus, when $\mathbf{F}_{\text{ext}} = \mathbf{0}$, the velocity $\mathbf{V}$ of the centre of mass remains constant (it moves uniformly in a straight line like a free particle), regardless of how complicated the internal motions or trajectories of the individual particles may be.

Classic Physical Illustrations of CM Motion:

- **Explosion of a Projectile in Mid-Air:** A shell fired along a parabolic trajectory explodes into fragments in mid-air. The explosion forces are purely internal and contribute nothing to the motion of the CM. Since the only external force (gravity) remains unchanged before and after the explosion, the centre of mass of the fragments continues along the exact same parabolic path it would have followed had there been no explosion.
- **Radioactive Decay of a Radium Nucleus:** A moving unstable radium (Ra) nucleus splits into a lighter radon (Rn) nucleus and an alpha particle (He nucleus) via internal nuclear forces. In the laboratory frame, the two product particles move in different directions such that their CM continues along the original straight-line path of the radium nucleus. In the *centre-of-mass reference frame* (where the CM is at rest), the radon nucleus and alpha particle simply fly apart back-to-back in opposite directions.
- **Binary (Double) Star System:** Two stars of equal mass interacting mutually via gravity (with no external forces) have a centre of mass C that moves with uniform velocity in a straight line. In the CM frame, the two stars move in a circular orbit about C, always remaining diametrically opposite to each other.



Tutorial Diagram 2: (a) Parabolic trajectory of the centre of mass is unaffected by an internal explosion. (b) Decay of a Radium nucleus into Radon and Helium in the laboratory frame and in the rest frame of the CM.

5. Vector Product, Angular Velocity, Torque & Angular Momentum

5.1 Vector (Cross) Product of Two Vectors

The vector product (or cross product) of two vectors $\mathbf{a}$ and $\mathbf{b}$, denoted $\mathbf{c} = \mathbf{a} \times \mathbf{b}$, is a vector with the following properties:

- **Magnitude:** $c = ab\sin\theta$, where a and b are the magnitudes of $\mathbf{a}$ and $\mathbf{b}$, and θ ($< 180^\circ$) is the smaller angle between them.
- **Direction:** $\mathbf{c}$ is perpendicular to the plane containing $\mathbf{a}$ and $\mathbf{b}$, determined by the **right-handed screw rule** (turning the screw head from $\mathbf{a}$ to $\mathbf{b}$ advances the tip along $\mathbf{c}$) or the **right-hand rule** (curling the fingers of the right hand from $\mathbf{a}$ to $\mathbf{b}$ points the stretched thumb along $\mathbf{c}$).
- **Non-Commutative (Anti-Commutative):** $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$ (hence $\mathbf{a} \times \mathbf{b} \neq \mathbf{b} \times \mathbf{a}$).
- **Distributive over Addition:** $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$.
- **Behaviour under Reflection (Parity):** Under plane mirror reflection ($x \to -x, y \to -y, z \to -z$), ordinary vectors reverse sign ($\mathbf{a} \to -\mathbf{a}$ and $\mathbf{b} \to -\mathbf{b}$), but their cross product does *not* change sign: $\mathbf{a} \times \mathbf{b} \to (-\mathbf{a}) \times (-\mathbf{b}) = \mathbf{a} \times \mathbf{b}$.
- **Self & Orthogonal Unit Vector Products:** $\mathbf{a} \times \mathbf{a} = \mathbf{0}$, so $\hat{\mathbf{i}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}} \times \hat{\mathbf{k}} = \mathbf{0}$. In cyclic order ($\hat{\mathbf{i}} \to \hat{\mathbf{j}} \to \hat{\mathbf{k}}$):
$$\hat{\mathbf{i}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}}, \quad \hat{\mathbf{j}} \times \hat{\mathbf{k}} = \hat{\mathbf{i}}, \quad \hat{\mathbf{k}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}} \quad \text{and} \quad \hat{\mathbf{j}} \times \hat{\mathbf{i}} = -\hat{\mathbf{k}}, \quad \hat{\mathbf{k}} \times \hat{\mathbf{j}} = -\hat{\mathbf{i}}, \quad \hat{\mathbf{i}} \times \hat{\mathbf{k}} = -\hat{\mathbf{j}}$$
- **Determinant & Component Form:**
$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = (a_y b_z - a_z b_y)\hat{\mathbf{i}} + (a_z b_x - a_x b_z)\hat{\mathbf{j}} + (a_x b_y - a_y b_x)\hat{\mathbf{k}}$$
- **Geometric Meaning:**
  - The area of a triangle contained between vectors $\mathbf{a}$ and $\mathbf{b}$ is $\frac{1}{2}|\mathbf{a} \times \mathbf{b}| = \frac{1}{2}ab\sin\theta$ (and the area of the parallelogram formed by them is $|\mathbf{a} \times \mathbf{b}|$).
  - The scalar triple product $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$ equals the volume of the parallelepiped formed on the three vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$.

5.2 Angular Velocity and Angular Acceleration

When a rigid body rotates about a fixed axis, every particle P_i at perpendicular distance r_i from the axis moves in a circle of radius r_i with the same instantaneous angular velocity $\omega = \frac{\mathrm{d}\theta}{\mathrm{d}t}$. Its linear speed is $v_i = \omega r_i$. Particles on the axis have $r_i = 0$ and therefore $v_i = 0$.

Angular velocity $\boldsymbol{\omega}$ is a vector directed along the axis of rotation according to the right-handed screw rule. If $\mathbf{r} = \mathbf{OP}$ is the position vector of particle P from an origin O on the axis, and C is the centre of the particle's circular path on the axis, then $\mathbf{r} = \mathbf{OC} + \mathbf{CP}$. Since $\boldsymbol{\omega}$ is parallel to $\mathbf{OC}$, $\boldsymbol{\omega} \times \mathbf{OC} = \mathbf{0}$, leaving $\boldsymbol{\omega} \times \mathbf{r} = \boldsymbol{\omega} \times \mathbf{CP}$, which has magnitude $\omega r_\perp$ and points tangentially along $\mathbf{v}$:

BOSONANCE

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} \qquad \text{and} \qquad \boldsymbol{\alpha} = \frac{\mathrm{d}\boldsymbol{\omega}}{\mathrm{d}t}$$

The relation $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$ holds both for fixed-axis rotation and for general rotation about a single fixed point O. For a fixed axis, the direction of $\boldsymbol{\omega}$ is fixed, so angular acceleration reduces to the scalar relation $\alpha = \frac{\mathrm{d}\omega}{\mathrm{d}t}$.

5.3 Moment of a Force (Torque) and Angular Momentum

Torque (Moment of Force): Torque is the rotational analogue of force. If a force $\mathbf{F}$ acts on a particle at position $\mathbf{r}$ relative to origin O, the torque $\boldsymbol{\tau}$ about O is:

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}, \qquad \tau = rF\sin\theta = r_\perp F = rF_\perp$$

where $r_\perp = r\sin\theta$ is the perpendicular distance of the line of action of $\mathbf{F}$ from O (moment arm), and $F_\perp = F\sin\theta$ is the component of $\mathbf{F}$ perpendicular to $\mathbf{r}$. Torque has dimensions $[\mathrm{ML^2T^{-2}}]$ and SI unit $\mathrm{N\,m}$. Although its dimensions are identical to work/energy, torque is a *vector* whereas work is a scalar; therefore, torque is written as $\mathrm{N\,m}$ and never as joule (J). Note that $\tau = 0$ if $r = 0$, $F = 0$, or $\theta = 0^\circ$ or 180° (line of action passes through the origin/hinge).

Angular Momentum (Moment of Momentum): For a single particle of mass m and linear momentum $\mathbf{p} = m\mathbf{v}$ at position $\mathbf{r}$ relative to O, its angular momentum $\mathbf{l}$ is:

$$\mathbf{l} = \mathbf{r} \times \mathbf{p}, \qquad l = rp\sin\theta = r_\perp p = rp_\perp$$

Differentiating $\mathbf{l} = \mathbf{r} \times \mathbf{p}$ with respect to time:

$$\frac{\mathrm{d}\mathbf{l}}{\mathrm{d}t} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} \times \mathbf{p} + \mathbf{r} \times \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = (\mathbf{v} \times m\mathbf{v}) + (\mathbf{r} \times \mathbf{F}) = \mathbf{0} + \boldsymbol{\tau} \implies \frac{\mathrm{d}\mathbf{l}}{\mathrm{d}t} = \boldsymbol{\tau}$$

Extension to a System of Particles: For a system of n particles, total angular momentum is $\mathbf{L} = \sum_{i=1}^{n} \mathbf{l}_i = \sum_{i=1}^{n} (\mathbf{r}_i \times \mathbf{p}_i)$. Differentiating with respect to time gives $\frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t} = \sum \boldsymbol{\tau}_i = \boldsymbol{\tau}_{\text{ext}} + \boldsymbol{\tau}_{\text{int}}$. Assuming Newton's third law *and* that the internal forces between any two particles act along the line joining the two particles, the net internal torque vanishes ($\boldsymbol{\tau}_{\text{int}} = \mathbf{0}$). Consequently:

$$\frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t} = \boldsymbol{\tau}_{\text{ext}}$$

Conservation of Angular Momentum: If the total external torque on a system is zero ($\boldsymbol{\tau}_{\text{ext}} = \mathbf{0}$), then $\frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t} = \mathbf{0}$, so $\mathbf{L} = \text{Constant}$ (meaning all three Cartesian components L_x, L_y, L_z remain constant).

Physical Demonstration — Precession of a Bicycle Rim: Suppose a bicycle wheel has its axle extended on both sides, with strings A and B tied to the two ends. If the wheel is not spinning and string B is released while holding string A, gravity creates an unbalanced torque and the rim immediately tilts downward. However, if the wheel is spun rapidly (giving it a large horizontal angular momentum $\mathbf{L}$ along the axle) and string B is released, the gravitational torque $\boldsymbol{\tau}$ (which is horizontal and perpendicular to both the vertical string A and the axle $\mathbf{L}$) causes the angular momentum vector $\mathbf{L}$ to rotate in a horizontal plane around string A. The spinning wheel remains vertical while its axle precesses around an axis perpendicular to both $\mathbf{L}$ and $\boldsymbol{\tau}$.

6. Equilibrium of a Rigid Body

A rigid body is in **mechanical equilibrium** if both its linear momentum $\mathbf{P}$ and angular momentum $\mathbf{L}$ are not changing with time (i.e., the body has neither linear acceleration nor angular acceleration). This requires two independent conditions:

1. **Translational Equilibrium:** The vector sum of all external forces is zero:
$$\sum_{i=1}^{n} \mathbf{F}_i = \mathbf{0} \quad \Longleftrightarrow \quad \sum F_{ix} = 0, \ \sum F_{iy} = 0, \ \sum F_{iz} = 0$$
2. **Rotational Equilibrium:** The vector sum of all external torques is zero:
$$\sum_{i=1}^{n} \boldsymbol{\tau}_i = \mathbf{0} \quad \Longleftrightarrow \quad \sum \tau_{ix} = 0, \ \sum \tau_{iy} = 0, \ \sum \tau_{iz} = 0$$

In general 3D problems, mechanical equilibrium requires **six** scalar conditions. For **coplanar forces** (all forces lying in the x-y plane), only **three** conditions are needed: $\sum F_x = 0$, $\sum F_y = 0$, and $\sum \tau_z = 0$. Furthermore, if translational equilibrium ($\sum \mathbf{F}_i = \mathbf{0}$) holds, the total torque on the rigid body is completely **independent of the choice of origin** about which torques are calculated.

6.1 Partial Equilibrium and Couple

A rigid body can be in translational equilibrium without rotational equilibrium, or vice versa:

- Consider a light rod AB of length $2a$ (midpoint C) with two equal parallel forces F acting in the *same* direction perpendicular to the rod at ends A and B. About C, the two moments aF are equal and opposite, so $\sum \boldsymbol{\tau} = \mathbf{0}$ (rotational equilibrium), but $\sum F = 2F \neq 0$ (not in translational equilibrium).
- Now reverse the force at B so that equal and opposite forces $\mathbf{F}$ and $-\mathbf{F}$ act perpendicular to the rod at A and B. Now $\sum \mathbf{F} = \mathbf{0}$ (translational equilibrium), but the two torques act in the same rotational sense, giving a net torque of magnitude $2aF \neq 0$ (not in rotational equilibrium).

Couple: A pair of forces of equal magnitude acting in opposite directions along different lines of action is called a **couple**. A couple produces *pure rotation without translation*. If forces $-\mathbf{F}$ and $+\mathbf{F}$ act at position vectors $\mathbf{r}_1$ (point A) and $\mathbf{r}_2$ (point B), the moment of the couple is:

$$\boldsymbol{\tau}_{\text{couple}} = \mathbf{r}_1 \times (-\mathbf{F}) + \mathbf{r}_2 \times \mathbf{F} = (\mathbf{r}_2 - \mathbf{r}_1) \times \mathbf{F} = \mathbf{AB} \times \mathbf{F}$$

This proves that the moment of a couple is independent of the origin O. Everyday examples include turning a bottle lid with two fingers and the torque exerted by the Earth's magnetic field on the north and south poles of a compass needle.

BOSONANCE

6.2 Principle of Moments (Ideal Lever)

An ideal lever is a light rod pivoted at a point called the **fulcrum** (like a playground see-saw or a beam balance). Let a load F_1 act at distance d_1 (*load arm*) and an effort F_2 act at distance d_2 (*effort arm*) on opposite sides of the fulcrum. If R is the upward support reaction at the fulcrum, translational equilibrium gives $R = F_1 + F_2$, and rotational equilibrium about the fulcrum gives:

$$d_1 F_1 - d_2 F_2 = 0 \implies d_1 F_1 = d_2 F_2 \quad (\text{Load arm} \times \text{Load} = \text{Effort arm} \times \text{Effort})$$

The **Mechanical Advantage (M.A.)** is defined as:

$$\text{M.A.} = \frac{F_1}{F_2} = \frac{d_2}{d_1}$$

If the effort arm d_2 is larger than the load arm d_1, then $\text{M.A.} > 1$, meaning a small effort can lift a large load.

6.3 Centre of Gravity (CG)

The **centre of gravity (CG)** of an extended body is defined as that point about which the total gravitational torque on the body is zero:

$$\boldsymbol{\tau}_g = \sum_{i=1}^{n} \mathbf{r}_i \times m_i \mathbf{g} = \mathbf{0}$$

If the body is small enough that the acceleration due to gravity $\mathbf{g}$ is uniform throughout the body, $\mathbf{g}$ can be factored out of the summation: $(\sum m_i \mathbf{r}_i) \times \mathbf{g} = \mathbf{0} \implies \sum m_i \mathbf{r}_i = \mathbf{0}$. Thus, **in a uniform gravitational field, the centre of gravity coincides with the centre of mass**. If a body is so extended that $\mathbf{g}$ varies from part to part, its CG and CM do not coincide. Experimentally, the CG of an irregular flat cardboard can be found by balancing it on a pencil tip, or by suspending it successively from two or three points (A, B, C) and finding the intersection of the vertical plumb lines.

7. Moment of Inertia, Rotational Kinematics & Dynamics

7.1 Moment of Inertia and Radius of Gyration

For a rigid body rotating about a fixed axis with angular speed ω, the ith particle (mass m_i, perpendicular distance r_i from the axis) has linear speed $v_i = r_i\omega$ and kinetic energy $k_i = \frac{1}{2} m_i v_i^2 = \frac{1}{2} m_i r_i^2 \omega^2$. Summing over all particles gives the total rotational kinetic energy K:

$$K = \frac{1}{2} I \omega^2 \qquad \text{where} \qquad I = \sum_{i=1}^{n} m_i r_i^2$$

The quantity I is the **moment of inertia** of the body about the given axis. It is the rotational analogue of mass M (rotational inertia) and has dimensions $[\text{ML}^2]$ and SI unit kg m^2. Unlike mass, I is not a fixed property of the body alone; it depends on the total mass, shape, size, distribution of mass, and the *position and orientation of the axis of rotation*. (Flywheels in engines are designed with a large moment of inertia to resist sudden changes in rotational speed and ensure a smooth ride.)

Radius of Gyration (k): If the entire mass M of the body were concentrated at a single point at perpendicular distance k from the axis such that it produced the exact same moment of inertia I, then:

$$I = Mk^2 \implies k = \sqrt{\frac{I}{M}}$$

#	Body (Homogeneous, Total Mass M)	Specified Axis of Rotation	Moment of Inertia (I)	Radius of Gyration (k)
1	Thin circular ring, radius R	Perpendicular to plane, at centre	MR^2	R
2	Thin circular ring, radius R	About any diameter	$\frac{1}{2}MR^2$	$\frac{R}{\sqrt{2}}$
3	Thin rod, length L	Perpendicular to rod, at midpoint	$\frac{1}{12}ML^2$	$\frac{L}{\sqrt{12}} = \frac{L}{2\sqrt{3}}$
4	Circular disc, radius R	Perpendicular to disc, at centre	$\frac{1}{2}MR^2$	$\frac{R}{\sqrt{2}}$
5	Circular disc, radius R	About any diameter	$\frac{1}{4}MR^2$	$\frac{R}{2}$
6	Hollow cylinder, radius R	Axis of cylinder	MR^2	R
7	Solid cylinder, radius R	Axis of cylinder	$\frac{1}{2}MR^2$	$\frac{R}{\sqrt{2}}$
8	Solid sphere, radius R	About any diameter	$\frac{2}{5}MR^2$	$\sqrt{\frac{2}{5}}R$

7.2 Kinematics and Dynamics of Rotation about a Fixed Axis

For rotation about a fixed axis (z-axis), only forces lying in planes perpendicular to the axis and position vector components perpendicular to the axis produce torque along the axis; perpendicular torque components are balanced by axis constraints. When a net axial torque τ rotates the body through $\mathrm{d}\theta$, the work done is $\mathrm{d}W = \tau\,\mathrm{d}\theta$ and instantaneous power is $P = \frac{\mathrm{d}W}{\mathrm{d}t} = \tau\omega$. Equating power to the rate of increase of rotational kinetic energy $\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}I\omega^2\right) = I\omega\alpha$ (for constant I) yields $\tau = I\alpha$.

Angular Momentum about a Fixed Axis: In general, for an individual particle, $\mathbf{l} = \mathbf{l}_z + \mathbf{OC} \times m\mathbf{v}$, so $\mathbf{l}$ is *not* parallel to $\boldsymbol{\omega}$. Summing over the rigid body gives $\mathbf{L} = \mathbf{L}_z + \mathbf{L}_\perp$, where $\mathbf{L}_z = I\omega\hat{\mathbf{k}}$. For bodies that are **symmetric about the axis of rotation**, diametrically opposite particle pairs cancel out $\mathbf{L}_\perp$ ($\mathbf{L}_\perp = \mathbf{0}$), so $\mathbf{L} = \mathbf{L}_z = I\omega\hat{\mathbf{k}}$. Differentiating gives $\frac{\mathrm{d}}{\mathrm{d}t}(I\omega) = \tau$. When the external axial torque is zero ($\tau_{\text{ext}} = 0$):

$$L_z = I\omega = \text{Constant} \implies I_1\omega_1 = I_2\omega_2$$

Application (Swivel Chair, Acrobat, Diver, Skater): When a person spinning on a frictionless swivel chair or turntable folds their outstretched arms inward, their moment of inertia I decreases ($I_2 < I_1$). By conservation of angular momentum ($I_1\omega_1 = I_2\omega_2$), their angular speed ω increases (

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$\omega_2 > \omega_1$). Notice that rotational kinetic energy $K = \frac{L^2}{2I}$ *increases* because the person performs internal muscular work in pulling their arms inward against centrifugal tendency!

Quantity	Linear (Translational) Motion	Rotational Motion about a Fixed Axis
Displacement	x	Angular displacement θ
Velocity	$v = \frac{\mathrm{d}x}{\mathrm{d}t}$	Angular velocity $\omega = \frac{\mathrm{d}\theta}{\mathrm{d}t}$
Acceleration	$a = \frac{\mathrm{d}v}{\mathrm{d}t}$	Angular acceleration $\alpha = \frac{\mathrm{d}\omega}{\mathrm{d}t}$
Kinematic Equations (Constant a or α)	$v = v_0 + at$ $x = x_0 + v_0 t + \frac{1}{2}at^2$ $v^2 = v_0^2 + 2a(x - x_0)$	$\omega = \omega_0 + \alpha t$ $\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$ $\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$
Inertia Parameter	Mass M	Moment of inertia I
Equation of Motion	$F = Ma$	$\tau = I\alpha$
Work Done	$\mathrm{d}W = F\,\mathrm{d}s$	$\mathrm{d}W = \tau\,\mathrm{d}\theta$
Kinetic Energy	$K = \frac{1}{2}Mv^2 = \frac{p^2}{2M}$	$K = \frac{1}{2}I\omega^2 = \frac{L^2}{2I}$
Instantaneous Power	$P = Fv$	$P = \tau\omega$
Momentum	$p = Mv$	$L_z = I\omega$ (for symmetric bodies, $L = I\omega$)

8. Comprehensive Worked Examples

Worked Example 1: Centre of Mass of Discrete Particles on an Equilateral Triangle

Problem: Find the centre of mass of three particles placed at the vertices of an equilateral triangle OAB of side length $0.5\,\mathrm{m}$. The masses of the particles at O, A, and B are $100\,\mathrm{g}$, $150\,\mathrm{g}$, and $200\,\mathrm{g}$ respectively.

Solution:

Choose vertex O as the origin $(0, 0)$ and side OA along the positive x-axis. Then the coordinates of the three vertices are:

- $O\,(m_1 = 100\,\mathrm{g}) : (x_1, y_1) = (0, 0)$
- $A\,(m_2 = 150\,\mathrm{g}) : (x_2, y_2) = (0.5\,\mathrm{m}, 0)$
- $B\,(m_3 = 200\,\mathrm{g}) : (x_3, y_3) = (0.5\cos 60^\circ, 0.5\sin 60^\circ) = (0.25\,\mathrm{m}, 0.25\sqrt{3}\,\mathrm{m})$

Total mass $M = 100 + 150 + 200 = 450\,\mathrm{g}$. Applying the centre of mass formulas:

$$X = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{M} = \frac{100(0) + 150(0.5) + 200(0.25)}{450} = \frac{75 + 50}{450} = \frac{125}{450}\,\mathrm{m} = \frac{5}{18}\,\mathrm{m}$$

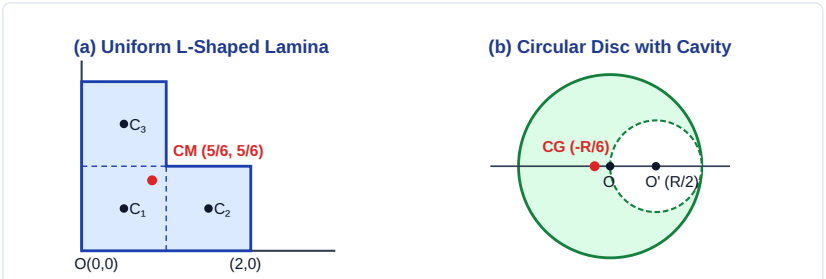
$$Y = \frac{m_1 y_1 + m_2 y_2 + m_3 y_3}{M} = \frac{100(0) + 150(0) + 200(0.25\sqrt{3})}{450} = \frac{50\sqrt{3}}{450}\,\mathrm{m} = \frac{\sqrt{3}}{9}\,\mathrm{m} = \frac{1}{3\sqrt{3}}\,\mathrm{m}$$

Thus, the centre of mass is at $\left(\frac{5}{18}\,\mathrm{m}, \frac{1}{3\sqrt{3}}\,\mathrm{m}\right)$. (Note: It does not coincide with the geometric centroid because the three masses are unequal.)

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Worked Example 2: Centre of Mass of Composite L-Shaped Lamina & Circular Cavity

Problem: (a) A uniform L-shaped thin lamina of total mass $3\,\mathrm{kg}$ is formed by joining three identical squares of side $1\,\mathrm{m}$ each: Square 1 covers $0 \le x \le 1, 0 \le y \le 1$; Square 2 covers $1 \le x \le 2, 0 \le y \le 1$; Square 3 covers $0 \le x \le 1, 1 \le y \le 2$. Find its centre of mass.
(b) From a uniform circular disc of radius R, a circular hole of radius $R/2$ is cut out such that the centre of the hole is at a distance $R/2$ from the centre of the original disc. Locate the centre of gravity of the resulting flat body.



Tutorial Diagram 3: (a) Composite L-shaped lamina divided into three squares. (b) Uniform disc of radius R with a circular hole of radius R/2 removed.

Solution:

(a) Since the lamina is uniform and has total mass $3\,\mathrm{kg}$, each $1\,\mathrm{m} \times 1\,\mathrm{m}$ square has mass $m_1 = m_2 = m_3 = 1\,\mathrm{kg}$. By symmetry, the centres of mass of the three squares are at their geometric centres: $C_1(1/2, 1/2)$, $C_2(3/2, 1/2)$, and $C_3(1/2, 3/2)$. Therefore:

$$X = \frac{1(1/2) + 1(3/2) + 1(1/2)}{1+1+1} = \frac{5}{6}\,\mathrm{m}, \qquad Y = \frac{1(1/2) + 1(1/2) + 1(3/2)}{1+1+1} = \frac{5}{6}\,\mathrm{m}$$

The CM lies at $\left(\frac{5}{6}\,\mathrm{m}, \frac{5}{6}\,\mathrm{m}\right)$, which lies on the line of symmetry $y = x$.

(b) Let the original complete disc of radius R have mass M and centre $O(0,0)$. Since mass is proportional to area (πR^2), the cutout circular disc of radius $R/2$ has area $\pi(R/2)^2 = \frac{1}{4}\pi R^2$ and therefore mass $m_{\text{cut}} = \frac{M}{4}$, centered at $x_{\text{cut}} = +\frac{R}{2}$. The remaining part has mass $m_{\text{rem}} = M - \frac{M}{4} = \frac{3M}{4}$ at position X_{rem}. Since the original disc is the superposition of the remaining part and the cutout disc, its CM is at $X_{\text{total}} = 0$:

$$0 = \frac{m_{\text{rem}} X_{\text{rem}} + m_{\text{cut}} x_{\text{cut}}}{M} \implies \left(\frac{3M}{4}\right) X_{\text{rem}} + \left(\frac{M}{4}\right)\left(\frac{R}{2}\right) = 0 \implies X_{\text{rem}} = -\frac{R}{6}$$

Thus, the centre of gravity (and CM) of the resulting flat body is at a distance $R/6$ from the centre O of the original disc, in the direction *opposite* to the centre of the cut-out hole.

Worked Example 3: Scalar Product, Vector Product & Torque Calculation

Problem: (a) Find the scalar and vector products of two vectors $\mathbf{a} = (3\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 5\hat{\mathbf{k}})$ and $\mathbf{b} = (-2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 3\hat{\mathbf{k}})$.
(b) Find the torque of a force $\mathbf{F} = (7\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 5\hat{\mathbf{k}})\,\mathrm{N}$ about the origin when the force acts on a particle whose position vector is $\mathbf{r} = (\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}})\,\mathrm{m}$.

Solution:

(a) The scalar (dot) product is:

$$\mathbf{a} \cdot \mathbf{b} = (3)(-2) + (-4)(1) + (5)(-3) = -6 - 4 - 15 = -25$$

The vector (cross) product is evaluated via determinant expansion:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 3 & -4 & 5 \\ -2 & 1 & -3 \end{vmatrix} = \hat{\mathbf{i}}\big((-4)(-3) - (5)(1)\big) - \hat{\mathbf{j}}\big((3)(-3) - (5)(-2)\big) + \hat{\mathbf{k}}\big((3)(1) - (-4)(-2)\big) = 7\hat{\mathbf{i}} - \hat{\mathbf{j}} - 5\hat{\mathbf{k}}$$

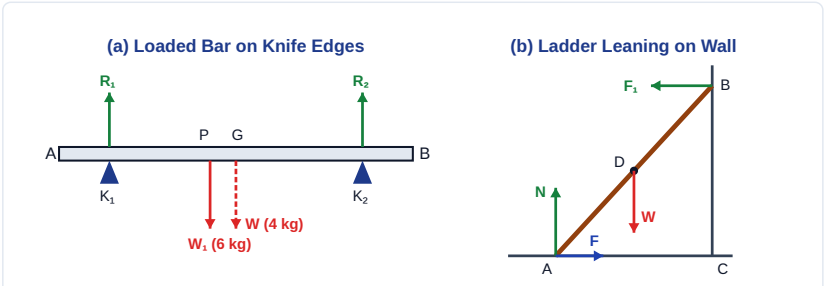
(b) Using $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$:

$$\boldsymbol{\tau} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & -1 & 1 \\ 7 & 3 & -5 \end{vmatrix} = \hat{\mathbf{i}}(5 - 3) - \hat{\mathbf{j}}(-5 - 7) + \hat{\mathbf{k}}(3 - (-7)) = (2\hat{\mathbf{i}} + 12\hat{\mathbf{j}} + 10\hat{\mathbf{k}})\,\mathrm{N\,m}$$

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Worked Example 4: Mechanical Equilibrium — Loaded Bar & Leaning Ladder

Problem: (a) A homogeneous metal bar $70\,\text{cm}$ long and $4.00\,\text{kg}$ in mass is supported on two knife-edges K_1 and K_2 placed $10\,\text{cm}$ from each end. A $6.00\,\text{kg}$ load is suspended at $30\,\text{cm}$ from end A. Find the normal reactions at the knife-edges. (Take $g = 9.8\,\text{m/s}^2$.)
(b) A $3\,\text{m}$ long uniform ladder weighing $20\,\text{kg}$ leans on a frictionless vertical wall with its feet resting on a rough floor $1\,\text{m}$ from the wall. Find the reaction forces of the wall and the floor.



Tutorial Diagram 4: Free-body force diagrams for (a) the supported metal bar and (b) the leaning ladder.

Solution:

(a) Here $AB = 70\,\text{cm}$, center of gravity G is at $AG = 35\,\text{cm}$, load $W_1 = 6.00g\,\text{N}$ is at $AP = 30\,\text{cm}$ (so $PG = 5\,\text{cm} = 0.05\,\text{m}$), and knife-edges are at $AK_1 = BK_2 = 10\,\text{cm}$ (so $K_1G = K_2G = 25\,\text{cm} = 0.25\,\text{m}$).

- Vertical translational equilibrium: $R_1 + R_2 = W + W_1 = (4.00 + 6.00) \times 9.8 = 98.00\,\text{N}$.
- Rotational equilibrium about G: $-R_1(K_1G) + W_1(PG) + R_2(K_2G) = 0 \implies -0.25R_1 + 0.05(6.00 \times 9.8) + 0.25R_2 = 0 \implies R_1 - R_2 = 1.2 \times 9.8 = 11.76\,\text{N}$.
- Solving the two equations: $R_1 = \frac{98.00 + 11.76}{2} = 54.88\,\text{N} \approx 55\,\text{N}$ and $R_2 = \frac{98.00 - 11.76}{2} = 43.12\,\text{N} \approx 43\,\text{N}$.

(b) For the ladder, $AB = 3\,\text{m}$ and $AC = 1\,\text{m}$, so by Pythagoras' theorem the height on the wall is $BC = \sqrt{3^2 - 1^2} = \sqrt{8} = 2\sqrt{2}\,\text{m}$. Weight $W = 20 \times 9.8 = 196.0\,\text{N}$ acts at midpoint D (horizontal distance $0.5\,\text{m}$ from A).

- Vertical equilibrium: $N - W = 0 \implies N = 196.0\,\text{N}$.
- Horizontal equilibrium: $F - F_1 = 0 \implies F = F_1$.
- Rotational equilibrium about foot A: $(2\sqrt{2})F_1 - (0.5)W = 0 \implies F_1 = \frac{W}{4\sqrt{2}} = \frac{196.0}{4\sqrt{2}} \approx 34.6\,\text{N}$.
- Therefore, wall reaction $F_1 = 34.6\,\text{N}$, friction $F = 34.6\,\text{N}$, and total floor reaction $F_2 = \sqrt{F^2 + N^2} = \sqrt{34.6^2 + 196.0^2} \approx 199.0\,\text{N}$ at angle $\alpha = \tan^{-1}(4\sqrt{2}) \approx 80^\circ$ with the horizontal.

Worked Example 5: Rotational Kinematics & Dynamics of a Flywheel

Problem: (a) The angular speed of a motor wheel is increased uniformly from $1200\,\text{rpm}$ to $3120\,\text{rpm}$ in $16\,\text{s}$. Find its angular acceleration and the number of revolutions completed during this time.
(b) A cord of negligible mass is wound around the rim of a flywheel (disc) of mass $M = 20\,\text{kg}$ and radius $R = 20\,\text{cm} = 0.20\,\text{m}$ mounted on a frictionless horizontal axle. A steady tangential pull of $F = 25\,\text{N}$ is applied to the cord. Find the angular acceleration α, the work done when $2\,\text{m}$ of cord is unwound, and the kinetic energy gained by the wheel starting from rest.

Solution:

(a) Convert rotational speeds from $\text{rpm}\ (\text{rev/min})$ to rad/s:

$$\omega_0 = \frac{2\pi \times 1200}{60} = 40\pi\,\text{rad/s}, \qquad \omega = \frac{2\pi \times 3120}{60} = 104\pi\,\text{rad/s}$$

$$\alpha = \frac{\omega - \omega_0}{t} = \frac{104\pi - 40\pi}{16} = 4\pi\,\text{rad/s}^2$$

Angular displacement $\theta = \omega_0 t + \frac{1}{2}\alpha t^2 = (40\pi)(16) + \frac{1}{2}(4\pi)(16)^2 = 640\pi + 512\pi = 1152\pi\,\text{rad}$. Number of revolutions $N = \frac{\theta}{2\pi} = \frac{1152\pi}{2\pi} = 576$.

(b) For the flywheel (uniform disc), moment of inertia $I = \frac{1}{2}MR^2 = \frac{1}{2}(20)(0.20)^2 = 0.4\,\text{kg m}^2$. Applied torque $\tau = FR = 25 \times 0.20 = 5.0\,\text{N m}$.

- Angular acceleration: $\alpha = \frac{\tau}{I} = \frac{5.0}{0.4} = 12.5\,\text{rad/s}^2$.
- Work done by the pull in unwinding $s = 2\,\text{m}$ of cord: $W = Fs = 25\,\text{N} \times 2\,\text{m} = 50\,\text{J}$.
- Angular displacement for $2\,\text{m}$ of cord: $\theta = \frac{s}{R} = \frac{2\,\text{m}}{0.20\,\text{m}} = 10\,\text{rad}$. From $\omega^2 = \omega_0^2 + 2\alpha\theta = 0 + 2(12.5)(10) = 250\,(\text{rad/s})^2$, the kinetic energy gained is $K = \frac{1}{2}I\omega^2 = \frac{1}{2}(0.4)(250) = 50\,\text{J}$, which matches the work done W by the work-energy theorem!

9. NEET UG Practice Questions with Worked Solutions

Note: The following original practice questions are strictly aligned with the NEET UG Physics examination pattern and syllabus.

Practice Question 1 (Single-Correct MCQ)

In the HCl molecule, the separation between the nuclei of the hydrogen and chlorine atoms is $1.27\,\text{Å}$ ($1\,\text{Å} = 10^{-10}\,\text{m}$). Given that a chlorine atom is 35.5 times as massive as a hydrogen atom and nearly all the mass of an atom is concentrated in its nucleus, what is the approximate distance of the centre of mass of the HCl molecule from the hydrogen nucleus?

(A) $0.035\,\text{Å}$

(B) $0.635\,\text{Å}$

(C) $1.24\,\text{Å}$

(D) $1.10\,\text{Å}$

Correct Answer: (C)

Worked Solution: Place the hydrogen nucleus of mass $m_1 = m$ at the origin $x_1 = 0$, and the chlorine nucleus of mass $m_2 = 35.5m$ at $x_2 = 1.27\,\text{Å}$. The position X of the centre of mass from the hydrogen nucleus is:

$$X = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} = \frac{m(0) + 35.5m(1.27\,\text{Å})}{m + 35.5m} = \frac{35.5 \times 1.27}{36.5}\,\text{Å} \approx 1.235\,\text{Å} \approx 1.24\,\text{Å}$$

Practice Question 2 (Single-Correct MCQ)

A child sits stationary at one end of a long trolley moving uniformly with speed V on a smooth, frictionless horizontal floor. If the child gets up and runs about on the trolley in any manner, what is the speed of the centre of mass of the (trolley + child) system?

(A) Increases if the child runs in the direction of motion of the trolley

(B) Decreases if the child runs opposite to the motion of the trolley

(C) Remains unchanged and equal to V

(D) Becomes zero as soon as the child starts running

Correct Answer: (C)

Worked Solution: Taking the (trolley + child) together as the system, the horizontal floor is smooth (frictionless), so there is no external horizontal force ($\mathbf{F}_{\text{ext}} = \mathbf{0}$) acting on the system. The forces exerted between the child's feet and the trolley are purely *internal forces*. By conservation of linear momentum, the velocity of the centre of mass remains constant at V.

Practice Question 3 (Single-Correct MCQ)

A non-uniform horizontal bar of length $2\,\text{m}$ and weight W is suspended at rest by two light strings attached to its ends. The strings at the left and right ends make angles of 36.9° and 53.1° respectively with the vertical ($\sin 36.9^\circ = \cos 53.1^\circ = 0.6$ and $\cos 36.9^\circ = \sin 53.1^\circ = 0.8$). What is the distance d of the centre of gravity of the bar from its left end?

(A) $0.72\,\text{m}$

(B) $1.00\,\text{m}$

(C) $1.28\,\text{m}$

(D) $0.64\,\text{m}$

Correct Answer: (A)

Worked Solution: Let T_1 and T_2 be the tensions in the left and right strings, where $\theta_1 = 36.9^\circ$ and $\theta_2 = 53.1^\circ$. For horizontal translational equilibrium:

$$T_1 \sin 36.9^\circ = T_2 \sin 53.1^\circ \implies 0.6\,T_1 = 0.8\,T_2 \implies T_2 = \frac{3}{4}T_1$$

For vertical equilibrium, $T_1 \cos 36.9^\circ + T_2 \cos 53.1^\circ = W$. Taking moments about the centre of gravity (located at distance d from the left end and $2 - d$ from the right end) for rotational equilibrium:

$$(T_1 \cos 36.9^\circ)\,d = (T_2 \cos 53.1^\circ)(2 - d)$$

$$(0.8\,T_1)\,d = \left(\frac{3}{4}T_1 \times 0.6\right)(2 - d) \implies 0.8\,d = 0.45(2 - d) \implies 1.25\,d = 0.90 \implies d = \frac{0.90}{1.25} = 0.72\,\text{m}$$

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Practice Question 4 (Single-Correct MCQ)

A car has a mass of $1800\,\text{kg}$. The distance between its front and back axles is $1.8\,\text{m}$, and its centre of gravity is $1.05\,\text{m}$ behind the front axle. Taking $g = 9.8\,\text{m/s}^2$, the normal force exerted by the level ground on *each* front wheel and *each* back wheel respectively are:

(A) $7350\,\text{N}$ and $10290\,\text{N}$

(B) $3675\,\text{N}$ and $5145\,\text{N}$

(C) $5145\,\text{N}$ and $3675\,\text{N}$

(D) $4410\,\text{N}$ and $4410\,\text{N}$

Correct Answer: (B)

Worked Solution: Let R_f and R_b be the total upward normal forces on the front axle (two front wheels) and back axle (two back wheels). Total weight $W = 1800 \times 9.8 = 17640\,\text{N}$. The CG is $1.05\,\text{m}$ from the front axle and $1.8 - 1.05 = 0.75\,\text{m}$ from the back axle. Taking moments about the CG:

$$R_f(1.05) = R_b(0.75) \implies R_b = \frac{1.05}{0.75}R_f = 1.4\,R_f$$

From $R_f + R_b = 17640\,\text{N}$, we get $2.4\,R_f = 17640 \implies R_f = 7350\,\text{N}$ and $R_b = 10290\,\text{N}$. Since there are two front wheels and two back wheels:

$$\text{Force on each front wheel} = \frac{7350}{2} = 3675\,\text{N}, \qquad \text{Force on each back wheel} = \frac{10290}{2} = 5145\,\text{N}$$

Practice Question 5 (Single-Correct MCQ)

A uniform metre stick ($100\,\text{cm}$ long) is balanced on a knife-edge at its centre ($50.0\,\text{cm}$ mark). When two coins, each of mass $5\,\text{g}$, are put one on top of the other at the $12.0\,\text{cm}$ mark, the stick is found to be balanced at the $45.0\,\text{cm}$ mark. What is the mass of the metre stick?

(A) $50.0\,\text{g}$

(B) $66.0\,\text{g}$

(C) $75.0\,\text{g}$

(D) $82.5\,\text{g}$

Correct Answer: (B)

Worked Solution: Since the metre stick initially balances at the $50.0\,\text{cm}$ mark, its weight Mg acts at the $50.0\,\text{cm}$ mark. The two coins have combined mass $m = 2 \times 5\,\text{g} = 10\,\text{g}$ at the $12.0\,\text{cm}$ mark. When balanced at the new fulcrum at $45.0\,\text{cm}$:

- Distance of coins from fulcrum: $d_1 = 45.0 - 12.0 = 33.0\,\text{cm}$.
- Distance of stick's CG from fulcrum: $d_2 = 50.0 - 45.0 = 5.0\,\text{cm}$.

Applying the principle of moments about the $45.0\,\text{cm}$ mark:

$$mgd_1 = Mgd_2 \implies (10\,\text{g})(33.0\,\text{cm}) = M(5.0\,\text{cm}) \implies M = \frac{330}{5.0} = 66.0\,\text{g}$$

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Practice Question 6 (Single-Correct MCQ)

Torques of equal magnitude τ are applied to a hollow cylinder and a solid sphere, both having the same mass M and radius R. The cylinder is free to rotate about its standard axis of symmetry, and the sphere is free to rotate about an axis passing through its centre. If both start from rest, what is the ratio of the angular speed of the solid sphere to that of the hollow cylinder ($\omega_{\text{sphere}} : \omega_{\text{cylinder}}$) after a given time t?

(A) $2 : 5$

(B) $5 : 2$

(C) $1 : 2$

(D) $4 : 5$

Correct Answer: (B)

Worked Solution: Moment of inertia of the hollow cylinder about its symmetry axis is $I_c = MR^2$, and that of the solid sphere about its diameter is $I_s = \frac{2}{5}MR^2$. Since both start from rest ($\omega_0 = 0$) under equal torque τ for time t:

$$\omega = \alpha t = \frac{\tau}{I} t \implies \omega \propto \frac{1}{I}$$

$$\frac{\omega_{\text{sphere}}}{\omega_{\text{cylinder}}} = \frac{I_c}{I_s} = \frac{MR^2}{\frac{2}{5}MR^2} = \frac{5}{2} \quad (\text{the solid sphere acquires a greater angular speed})$$

Practice Question 7 (Single-Correct MCQ)

A solid cylinder of mass $20\,\text{kg}$ and radius $0.25\,\text{m}$ rotates about its symmetry axis with an angular speed of $100\,\text{rad}\,\text{s}^{-1}$. What are the rotational kinetic energy and the magnitude of angular momentum of the cylinder about its axis?

(A) $3125\,\text{J}$ and $62.5\,\text{J}\,\text{s}$

(B) $6250\,\text{J}$ and $125.0\,\text{J}\,\text{s}$

(C) $3125\,\text{J}$ and $125.0\,\text{J}\,\text{s}$

(D) $1562.5\,\text{J}$ and $62.5\,\text{J}\,\text{s}$

Correct Answer: (A)

Worked Solution: For a solid cylinder about its axis, moment of inertia is:

$$I = \frac{1}{2}MR^2 = \frac{1}{2}(20\,\text{kg})(0.25\,\text{m})^2 = 10 \times 0.0625 = 0.625\,\text{kg}\,\text{m}^2$$

Therefore, the rotational kinetic energy K and angular momentum L are:

$$K = \frac{1}{2}I\omega^2 = \frac{1}{2}(0.625)(100)^2 = 3125\,\text{J}, \qquad L = I\omega = (0.625)(100) = 62.5\,\text{J}\,\text{s}$$

BOSONANCE

Practice Question 8 (Single-Correct MCQ)

A child stands at the centre of a frictionless turntable with his two arms outstretched. The turntable is set rotating with an angular speed of $40\,\text{rev/min}$. When the child folds his hands back, his moment of inertia reduces to $\frac{2}{5}$ times the initial value. What are (i) the new angular speed of the child, and (ii) the ratio of his final rotational kinetic energy to his initial rotational kinetic energy (K_f/K_i)?

(A) $100\,\text{rev/min}$ and $5:2$

(B) $16\,\text{rev/min}$ and $2:5$

(C) $100\,\text{rev/min}$ and $1:1$

(D) $80\,\text{rev/min}$ and $2:1$

Correct Answer: (A)

Worked Solution: Since the turntable rotates without friction, external torque $\tau_{\text{ext}} = 0$, so angular momentum is conserved ($I_i\omega_i = I_f\omega_f$). Given $I_f = \frac{2}{5}I_i$ and $\omega_i = 40\,\text{rev/min}$:

$$\omega_f = \frac{I_i}{I_f}\omega_i = \frac{5}{2} \times 40 = 100\,\text{rev/min}$$

The ratio of rotational kinetic energies is:

$$\frac{K_f}{K_i} = \frac{\frac{1}{2}I_f\omega_f^2}{\frac{1}{2}I_i\omega_i^2} = \left(\frac{2}{5}\right)\left(\frac{5}{2}\right)^2 = \frac{5}{2} = 2.5$$

(This increase in kinetic energy comes from the internal muscular energy expended by the child in pulling his arms inward.)

Practice Question 9 (Single-Correct MCQ)

A rope of negligible mass is wound round a hollow cylinder of mass $3\,\text{kg}$ and radius $40\,\text{cm}$. If the rope is pulled tangentially with a force of $30\,\text{N}$ without slipping, what are the angular acceleration of the cylinder and the linear acceleration of the rope?

(A) $12.5\,\text{rad}\,\text{s}^{-2}$ and $5\,\text{m}\,\text{s}^{-2}$

(B) $25\,\text{rad}\,\text{s}^{-2}$ and $10\,\text{m}\,\text{s}^{-2}$

(C) $50\,\text{rad}\,\text{s}^{-2}$ and $20\,\text{m}\,\text{s}^{-2}$

(D) $10\,\text{rad}\,\text{s}^{-2}$ and $4\,\text{m}\,\text{s}^{-2}$

Correct Answer: (B)

Worked Solution: For a hollow cylinder of mass $M = 3\,\text{kg}$ and radius $R = 0.40\,\text{m}$, the moment of inertia is $I = MR^2 = 3 \times (0.40)^2 = 0.48\,\text{kg}\,\text{m}^2$. Torque $\tau = FR = 30 \times 0.40 = 12\,\text{N}\,\text{m}$. Therefore:

$$\alpha = \frac{\tau}{I} = \frac{12\,\text{N}\,\text{m}}{0.48\,\text{kg}\,\text{m}^2} = 25\,\text{rad}\,\text{s}^{-2}$$

Since there is no slipping, the linear acceleration of the rope is $a = R\alpha = 0.40\,\text{m} \times 25\,\text{rad}\,\text{s}^{-2} = 10\,\text{m}\,\text{s}^{-2}$.

Practice Question 10 (Single-Correct MCQ)

To maintain a rotor at a uniform angular speed of $200\,\text{rad}\,\text{s}^{-1}$ against frictional losses, an engine needs to transmit a steady torque of $180\,\text{N}\,\text{m}$. Assuming the engine is 100% efficient, what is the power required by the engine?

(A) $18\,\text{kW}$

(B) $36\,\text{kW}$

(C) $9\,\text{kW}$

(D) $72\,\text{kW}$

Correct Answer: (B)

Worked Solution: Power transmitted in rotational motion is given by:

$$P = \tau\omega = (180\,\text{N}\,\text{m}) \times (200\,\text{rad}\,\text{s}^{-1}) = 36000\,\text{W} = 36\,\text{kW}$$

BOSONANCE

Practice Question 11 (Single-Correct MCQ)

An oxygen molecule has a mass of $5.30 \times 10^{-26}\,\text{kg}$ and a moment of inertia of $1.94 \times 10^{-46}\,\text{kg}\,\text{m}^2$ about an axis through its centre perpendicular to the line joining the two atoms. Suppose the mean speed of such a molecule in a gas is $500\,\text{m/s}$ and that its kinetic energy of rotation is two-thirds of its kinetic energy of translation. What is the average angular velocity of the molecule?

(A) $3.40 \times 10^{12}\,\text{rad/s}$

(B) $6.75 \times 10^{12}\,\text{rad/s}$

(C) $1.35 \times 10^{13}\,\text{rad/s}$

(D) $4.56 \times 10^{11}\,\text{rad/s}$

Correct Answer: (B)

Worked Solution: Given $K_{\text{rot}} = \frac{2}{3}K_{\text{trans}}$, we have:

$$\frac{1}{2}I\omega^2 = \frac{2}{3}\left(\frac{1}{2}mv^2\right) \implies \omega = \sqrt{\frac{2mv^2}{3I}} = v\sqrt{\frac{2m}{3I}}$$

Substituting $m = 5.30 \times 10^{-26}\,\text{kg}$, $I = 1.94 \times 10^{-46}\,\text{kg}\,\text{m}^2$, and $v = 500\,\text{m/s}$:

$$\omega = 500 \times \sqrt{\frac{2 \times 5.30 \times 10^{-26}}{3 \times 1.94 \times 10^{-46}}} = 500 \times \sqrt{1.821 \times 10^{20}} \approx 500 \times 1.35 \times 10^{10} = 6.75 \times 10^{12}\,\text{rad/s}$$

Practice Question 12 (Multi-Statement Evaluation)

Consider the following two statements regarding angular momentum of particles:
Statement I: If a particle moves only in the x-y plane ($z = 0$ and $p_z = 0$), its angular momentum vector $\mathbf{l} = \mathbf{r} \times \mathbf{p}$ about the origin has only a z-component given by $l_z = xp_y - yp_x$.
Statement II: Two particles, each of mass m and speed v, travel in opposite directions along parallel lines separated by a perpendicular distance d. The total angular momentum vector of this two-particle system has magnitude mvd and is the same regardless of the point about which the angular momentum is taken.
In the light of the above statements, choose the correct option:

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true, but Statement II is false.

(D) Statement I is false, but Statement II is true.

Correct Answer: (A)

Worked Solution:

For Statement I: $\mathbf{l} = (yp_z - zp_y)\hat{\mathbf{i}} + (zp_x - xp_z)\hat{\mathbf{j}} + (xp_y - yp_x)\hat{\mathbf{k}}$. When $z = 0$ and $p_z = 0$, $l_x = 0$ and $l_y = 0$, leaving $\mathbf{l} = (xp_y - yp_x)\hat{\mathbf{k}}$. Thus Statement I is true.

For Statement II: The two particles have momenta $\mathbf{p}_1 = m\mathbf{v}$ and $\mathbf{p}_2 = -m\mathbf{v}$ (so total linear momentum $\mathbf{P} = \mathbf{0}$). About any arbitrary origin O, $\mathbf{L} = \mathbf{r}_1 \times m\mathbf{v} + \mathbf{r}_2 \times (-m\mathbf{v}) = (\mathbf{r}_1 - \mathbf{r}_2) \times m\mathbf{v}$, which has magnitude mvd and is independent of the choice of O. Thus Statement II is also true.

BOSONANCE

Practice Question 13 (Assertion–Reason Format)

Assertion (A): For a rigid body in translational equilibrium, the net external torque acting on the body is the same about any point chosen as the origin.
Reason (R): Whenever the net external force on a rigid body is zero, the net external torque on the body must also be zero.
Choose the correct option:

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (C)

Worked Solution: If the origin is shifted by $\mathbf{r}_0$, the new position vectors are $\mathbf{r}_i' = \mathbf{r}_i - \mathbf{r}_0$, so the new torque is $\boldsymbol{\tau}' = \sum(\mathbf{r}_i - \mathbf{r}_0) \times \mathbf{F}_i = \boldsymbol{\tau} - \mathbf{r}_0 \times (\sum \mathbf{F}_i)$. In translational equilibrium ($\sum \mathbf{F}_i = \mathbf{0}$), $\boldsymbol{\tau}' = \boldsymbol{\tau}$, so Assertion (A) is true. However, Reason (R) is false because $\sum \mathbf{F}_i = \mathbf{0}$ and $\sum \boldsymbol{\tau}_i = \mathbf{0}$ are independent conditions (for example, in a couple, net external force is zero while net torque is non-zero).

10. Common Pitfalls & Quick Revision Checklist

High-Yield NEET Pitfalls to Avoid:

- **Confusing Torque and Work Units:** Both torque ($\mathbf{r} \times \mathbf{F}$) and work ($\mathbf{F} \cdot \mathbf{s}$) have dimensions $[\mathrm{ML^2T^{-2}}]$, but torque is a vector ($\mathrm{N\,m}$) while work is a scalar (J). Never express torque in joules.
- **Assuming $\mathbf{L}$ and $\boldsymbol{\omega}$ are Always Parallel:** Unlike $\mathbf{p} = m\mathbf{v}$ (where $\mathbf{p}$ is always parallel to $\mathbf{v}$), angular momentum $\mathbf{L} = \mathbf{L}_z + \mathbf{L}_\perp$ is parallel to $\boldsymbol{\omega}$ ($\mathbf{L} = I\boldsymbol{\omega}$) *only* when the fixed axis of rotation is an axis of symmetry of the rigid body.
- **Equating CM and CG Universally:** The centre of mass depends solely on mass distribution, whereas the centre of gravity is where net gravitational torque is zero. They coincide *only* in a uniform gravitational field.
- **Forgetting Kinetic Energy Separation:** Total kinetic energy of a system of particles is $K = K' + \frac{1}{2}MV^2$, where K' is the kinetic energy of motion relative to the CM and $\frac{1}{2}MV^2$ is the translational kinetic energy of the CM.
- **Confusing Ring/Disc Axes:** Check carefully whether the axis is *perpendicular to the plane at the centre* ($I_{\text{ring}} = MR^2$, $I_{\text{disc}} = \frac{1}{2}MR^2$) or *along a diameter* ($I_{\text{ring}} = \frac{1}{2}MR^2$, $I_{\text{disc}} = \frac{1}{4}MR^2$).

10.1 Final Recap of Key Quantities

Physical Quantity	Symbol	Dimensions	SI Unit	Defining Relation
Position of Centre of Mass	$\mathbf{R}$	$[\mathrm{L}]$	m	$\mathbf{R} = \frac{1}{M}\sum m_i\mathbf{r}_i = \frac{1}{M}\int \mathbf{r}\,dm$
Angular Velocity	$\boldsymbol{\omega}$	$[\mathrm{T^{-1}}]$	$\mathrm{rad\,s^{-1}}$	$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$, $\omega = \frac{d\theta}{dt}$
Torque (Moment of Force)	$\boldsymbol{\tau}$	$[\mathrm{ML^2T^{-2}}]$	$\mathrm{N\,m}$	$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$, $\tau_z = I\alpha$
Angular Momentum	$\mathbf{L}$	$[\mathrm{ML^2T^{-1}}]$	$\mathrm{J\,s}$ or $\mathrm{kg\,m^2\,s^{-1}}$	$\mathbf{L} = \sum \mathbf{r}_i \times \mathbf{p}_i$, $L_z = I\omega$
Moment of Inertia	I	$[\mathrm{ML^2}]$	$\mathrm{kg\,m^2}$	$I = \sum m_i r_i^2 = Mk^2$

BOSONANCE